

[DC] Maximizing Natural Walking in Virtual Environments

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ABSTRACT

Being restricted to small physical spaces limits the practicability of real walking in extensive virtual areas. Thus, teleportation or other locomotion metaphors are needed to fully navigate such virtual spaces. Nevertheless, since natural walking reflects the naturalness and intuitiveness of human locomotion well, it is still considered the best locomotion method. Therefore, researchers have mainly focused on developing concepts to increase the applicability of real walking in virtual spaces exceeding the physical space availability. These redirection methods mainly consist of overlapping spaces, redirected walking, and change blindness. Some of these methods have been extended to environments with multiple users. However, most literature considers only one method at the time applied to single users. Since this work focuses on the maximization of the natural walking experience, the combination of RDW in overlapping spaces with optimized reset strategies may complement and thus improve the final user experience. Further, most of previous approaches presume the boundaries of the physical space to be known beforehand and that this space is convex and free of obstacles. In my Ph.D. work, I will first look into combining the existing redirection methods in a known tracking space to maximize the walking distances that multi-users can navigate without colliding. In a subsequent step, I will extend these methods to an unknown and non-convex physical space. To discover and thus expand the available physical space, I will look into various exploration strategies that allow efficient space discovery. Developing the two previous ideas would only reduce the collision risk, accordingly, resets are still necessary. Hence, I will further investigate new reset strategies based on path prediction that allow potentially resetting users in early stages before they are even close to obstacles or boundaries, strategically reducing future resets.

Index Terms: Locomotion—Redirection techniques—Redirected walking—Natural Walking

1 INTRODUCTION

Real walking in virtual environments (VEs) can significantly increase immersion, but comes with the problem that VEs are typically larger than the physical environment (PE). In that case, a user wearing a head-mounted display (HMD) blocking the line of sight to reality could end up colliding with the boundaries or obstacles of the physical space. Early approaches that allowed safe locomotion closely imitating real walking were so-called repositioning systems. Most of them regrettably never leave research because they fail to fully evoke the sensation of real walking [8]. Another set of methods focuses on steering the user away from boundaries. These methods are called redirection techniques and a taxonomy to classify them has been proposed [5]. In my Ph.D. work, I will focus on three algorithms: Redirected Walking (RDW), Change Blindness, and Self-overlapping spaces (SOS).

Redirected Walking is a technique that exploits that human visual perception outweighs our other senses such as proprioception

or audio. Instead of mapping the user's movements one-to-one into the VE, these movements are slightly distorted, deviating the virtual from the real trajectory [3]. Such so-called redirection techniques can be employed with varying intensity, and if applied below perception thresholds tend to go unnoticed, without invoking simulator sickness [4].

Change Blindness is a redirection method that manipulates parts of the VE, leaving the user unaware of those changes. For example, a door they entered a room through moves to a different location while the user is occupied within the room. While this expands the available virtual space, it can irritate the user and is harder to generalize to arbitrary spaces.

Self-overlapping Spaces is a redirection method that refers to having separate interconnected areas of a virtual space occupying the same physical space. Users are usually led from one area to the next through corridors or an elevator ride [9] to hide the overlapping structure of these spaces.

Even though redirection techniques are applicable while users are exploring a VE, a user will inevitably attempt at some point to cross the boundary of the tracking space. Crucially, a safety measure is needed to prevent the user from such a collision with the tracking boundaries. The most common one forms so-called resets [10], where the user is stopped from further advancing by a visual prompt, which requires the user to fulfill a specific task before the virtual exploration is continued.

Virtual reality applications often involve multiple users, sometimes in co-located tracking spaces. Besides inherently rendering the tracking space dynamic and non-convex, users can further also collide with each other. So far, such multi-user applications were only investigated regarding basic RDW concepts including respective resets [7]. This was however never extended to the other mentioned redirection techniques of change blindness and SOS.

The goal of this Ph.D. thesis is to maximize the possible VE where the user can navigate by real walking, thus allowing uninterrupted immersion. The work will initially focus on finding novel ways to combine the three redirection techniques previously mentioned. In the next step, exploring VEs in unknown PEs with dynamic boundaries will be tackled. This would allow a generalization of the methods to new shapes of spaces while removing previous calibration in which the user defines the boundaries of the tracking space prior to virtual exploration. Finally, in the third step, I will look into optimizing resets since these still occur even though redirection is applied.

2 COMBINING DIFFERENT REDIRECTION TECHNIQUES

Researchers have developed various methods allowing users to walk in a VE larger than the physical space. These approaches include RDW, Change Blindness, and SOS. However, most of the work mainly focused on using a single method at a time. In my Ph.D. work, the goal is to develop an accessible locomotion library that allows using multiple methods simultaneously, this would be the first milestone. The library follows a similar structure to previously presented RDW libraries [2] with the addition of allowing SOS and Change Blindness. In the next stage, I aim to pair the methods and test user acceptance in brief pilot user studies, before eventually combining all three methods in a more extensive approach. In this part of the work, I also investigate the requirements of the VEs

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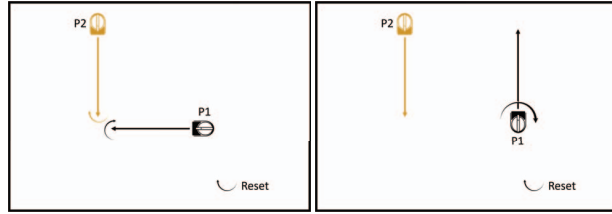


Figure 1: Illustration of the envisioned predictive reset technique: Without early reset (left); Early reset triggered (Right)

and PEs that makes each redirection method and their combination more beneficial, i.e., when alignment-based RDW [12] performs better than S2G. One of the hypotheses that would be tested is that alignment-based RDW can improve SOS techniques.

3 WALKING IN AN UNKNOWN SPACE

Previous work in RDW mainly relied on the assumption that the tracking space is known and empty. Relaxing these assumptions would simplify the usage of RDW in commercial applications, but requires a more elaborate algorithmic approach. A method has been proposed to map and integrate the environment into a RDW algorithm [1]. Other researchers are looking into this problem from a path-planning perspective [11]. This Ph.D. work however suggests to treat the prevalent issue as an optimization problem. The algorithm must choose between exploring and exploiting available information on the environment. Crucially, a steering policy follows an optimality principle that redirects the user while aggressively pushing toward unknown boundaries and extending the known area of the given tracking space before switching to a more conservative approach when satisfied. The goal is to maximize the exploration of walkable spaces that can be navigated by the user. My work will consist of three main contributions. The first one is testing different representations for unknown areas. For example, if APFs are used, then these areas could be repulsive or attractive. The other contribution would be to find requirements and properties compelling the algorithm to optimize when to push the user toward such spaces. Finally, once the space has been explored, the algorithm automatically outlines an overlapping strategy of the VE based on the available space, maximizing the discrete walkable area of both worlds.

4 RESET OPTIMIZATION

As long as the VE is larger than the PE, the user is always endangered to collide with physical boundaries or other users. Thus, resets are still important as a safety measure, always ensuring a collision-free VR experience for the user. Various strategies have been implemented, some are based on predicting the user's next steps, such as the SFR2G [6], others focus on resets that align the PE and VE [12]. In this work, I will delve deeper into resets and elaborate new methods exploring optimization strategies while aligning with the combination of redirection methods previously introduced. The two ideas I want to look into are: In multi-user environments, users often reset caused by each other. Currently, both users simultaneously enter a reset state, which apparently impairs the experience of both. Consequently, one could predict the trajectory of the users and trigger a reset in an earlier stage for a particular user, preventing the need to reset both users as shown in Fig. 1. The same could be applied to a single user where predicting their trajectory can allow a reset and avoid two later resets near the boundaries.

5 CONCLUSION

In this consortium, I present the ideas that I will examine in my Ph.D. work. These ideas aim to allow users to navigate VEs by

walking in constrained PEs. First, I will investigate an adequate way to combine the redirection methods and identify possible VEs and PEs combinations where they can be beneficial. Next, I will delve deeper into RDW in an unknown space allowing to relax the constraint of needing known physical space boundaries. Thirdly, I acknowledge that resets are needed when a collision is inevitable. Thus, I will look into optimizing these reset strategies. A library will be developed to integrate the redirection methods more efficiently. Then, an algorithm will be created which automatically assigns the best method for each combination of PE and VE. RDW in an unknown space will be considered an optimization problem where the algorithm needs to exploit its information versus exploring new spaces. In the last step, I look into possible reset strategies already implemented to determine the relations between an optimal strategy adjusted to the given PE.

REFERENCES

- [1] C. Hirt, M. Zank, and A. Kunz. Geometry extraction for ad hoc redirected walking using a slam device. In *International Conference on Augmented Reality, Virtual Reality and Computer Graphics*, pp. 35–53. Springer, 2018.
- [2] Y.-J. Li, M. Wang, F. Steinicke, and Q. Zhao. Openrdw: A redirected walking library and benchmark with multi-user, learning-based functionalities and state-of-the-art algorithms. In *2021 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pp. 21–30, 2021. doi: 10.1109/ISMAR52148.2021.00016
- [3] S. Razzaque. *Redirected walking*. The University of North Carolina at Chapel Hill, 2005.
- [4] F. Steinicke, G. Bruder, J. Jerald, H. Frenz, and M. Lappe. Estimation of detection thresholds for redirected walking techniques. *IEEE Transactions on Visualization and Computer Graphics*, 16(1):17–27, 2010. doi: 10.1109/TVCG.2009.62
- [5] E. A. Suma, G. Bruder, F. Steinicke, D. M. Krum, and M. Bolas. A taxonomy for deploying redirection techniques in immersive virtual environments. In *2012 IEEE Virtual Reality Workshops (VRW)*, pp. 43–46, 2012. doi: 10.1109/VR.2012.6180877
- [6] J. Thomas and E. Rosenberg. A general reactive algorithm for redirected walking using artificial potential functions. In *IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 56–62, 03 2019. doi: 10.1109/VR.2019.8797983
- [7] J. Thomas and E. S. Rosenberg. A general reactive algorithm for redirected walking using artificial potential functions. In *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 56–62, 2019. doi: 10.1109/VR.2019.8797983
- [8] M. Usoh, K. Arthur, M. C. Whitton, R. Bastos, A. Steed, M. Slater, and F. P. Brooks. Walking >walking-in-place>flying, in virtual environments. In *Proceedings of the 26th Annual Conference on Computer Graphics and Interactive Techniques, SIGGRAPH '99*, p. 359–364. ACM Press/Addison-Wesley Publishing Co., USA, 1999. doi: 10.1145/311535.311589
- [9] K. Vasylevska and H. Kaufmann. Compressing vr: Fitting large virtual environments within limited physical space. *IEEE computer graphics and applications*, 37(5):85–91, 2017.
- [10] B. Williams, G. Narasimham, B. Rump, T. P. McNamara, T. H. Carr, J. Rieser, and B. Bodenheimer. Exploring large virtual environments with an hmd when physical space is limited. In *Proceedings of the 4th Symposium on Applied Perception in Graphics and Visualization, APGV '07*, p. 41–48. Association for Computing Machinery, New York, NY, USA, 2007. doi: 10.1145/1272582.1272590
- [11] N. L. Williams. Robust redirected walking in the wild. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 922–923, 2022. doi: 10.1109/VRW55335.2022.00312
- [12] N. L. Williams, A. Bera, and D. Manocha. Redirected walking in static and dynamic scenes using visibility polygons. *CoRR*, abs/2106.06807, 2021.